Zachary L. Steel

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EDUCATION

University of California, Davis, Ph.D. Ecology 2014–2018

University of California, Davis, M.S. Ecology 2008–2011

Whitman College, B.A. Biology 2002–2006

EMPLOYMENT

USDA Forest Service Rocky Mountain Research Station:

Research Ecologist 2022–Present

University of California, Berkeley:

Post-doctoral Researcher 2019–Present

University of California, Davis:

Graduate Student Researcher 2014–2018

Jr. Specialist Researcher 2012–2014

Fire Science Consortium Coordinator 2012–2013

US Fulbright Program, Santiago, Chile:

Research Fellow 2011

University of California, Davis:

Graduate Student Researcher 2008–2011

University of California, Santa Cruz:

Assistant Biologist 2007–2008

PRBO Conservation Sciences:

Biology Technician 2007–2008

PUBLICATIONS

Journal Articles

In Press – Lather, E., K. Carroll, A. Carlson, D. Helmers, M. H. Mockrin, **Z. L. Steel**, V. C. Radeloff, and A. M. Pidgeon. "Areas with high predicted breeding bird richness lack protection and are likely affected by buildings in their vicinity". *Ecological Applications*

2026 – Mills, K., M. Leclerc, M. Ditmer, D. Stoner, **Z. L. Steel**, J. Sexton, P. Wang, K. Hersey, C. Schroeder, A. Andreasen, K. Engebretsen, J. Young, B. McMillan, R. Larsen, D. Hall, B. Holton, D. Choate, and N. Carter. "Burned habitats facilitate predation and lead to long-term reductions in prey success". *Journal of Animal Ecology*.

2026 – Parlin, A., M. Ditmer, **Z. L. Steel**, J. Alder, K. Hersey, V. Stout, W. Alexander, B. McMillan, R. Larsen, M. DeVivo, A. Townsend, J. Cain, M. Peyton, B. Gardner, G. Wittemyer. "Smoke Signals: Measuring Wildlife Smoke Exposure with Animal-Borne and Stationary Sensors". *Methods in Ecology and Evolution*.

2026 – Van den Bosch, M., H. Boone, J. Costanza, R. Peek, and **Z. L. Steel**. "Long-term drought exposure shapes declines and range shifts in aridland birds". *Conservation Letters*: 19(3):e70069.

2026 – McTigue, L. E., M. van den Bosch, H. M. McCaslin, L. N. Jones, H. M. Boone, J. M. Mola, **Z. L. Steel**. "Global trends in drought impacts on Vertebrates - a review". *Global Change Biology*: 32(2):e70752.

2026 – Mills, K., M. Leclerc, M. Ditmer, **Z. L. Steel**, D. Stoner, J. Sexton, P. Wang, K. Hersey, D. DeBloois, C. Schroeder, J. Young, A. Andreasen, K. Longshore, P. Jackson, D. Hall, K. Engebretsen, and N. Carter. "Long-term benefits of burns for large mammal habitat undermined by large, severe fires in the American West". *Ecography*: 2:e08225.

2025 – Nesbit, K. A., B. M. Collins, **Z. L. Steel**, J. J. Battles, M. L. Goulden, and S. L. Stephens. "Multi-scale assessment of wildfire use impacts on carbon stocks in the Sierra Nevada, CA". *Fire Ecology*: 21:49.

2025 – van den Bosch, M.[1], **Z. L. Steel**[1], J. K. Costanza, K. F. Kellner, and R. A. Peek. "Increased drought drives avian community declines in the warm deserts of the United States". *Biological Conservation*: 210:111393.
[1]*Authors contributed equally to this work*

2025 – Calhoun, K., P. Parker-Shames, **Z. Steel**, H. Oyler, and J. Brashares. "Severity outweighs pyrodiversity in shaping avian and bat species distributions following an oak woodland megafire". *Ecosphere*: e70263.

2025 – Shive, K. L., C. A. Knight, **Z. L. Steel**, C. K. Stanley, and K. N. Wilson. "Leveraging wildfire to augment forest management and amplify forest resilience". *Ecosphere*: 16(6):e70306.

2025 – van den Bosch, M., J. K. Costanza, R. A. Peek, and **Z. L. Steel**. "Projected increases in climate extremes across global vertebrate diversity hotspots". *Global Change Biology*: 31(6):e70272.

2025 – **Steel, Z. L.**, A. M. Fogg, R. Floreani Buzbee, K. Wilkin, B. M. Collins, R. Burnett, M. D. Meyer, A. Wuenschel, and S. L. Stephens. "Finding floral and faunal species richness optima among active fire regimes". *Conservation Biology*:e70079.

2025 – Jones, G. M., L. E. Hankin, R. Hart, M. D. Meyer, J. Regelbrugge, **Z. L. Steel**, and C. Thompson. "Collapse and restoration of mature forest habitat in California". *Biological Conservation*: 308:111241.

2025 – Miller-ter Kuile, A., J. Sanderlin, J. Ayars, H. Chmura, M. Dressen, J. Golding, G. Jones, R. Kirby, K. Norman, **Z. L. Steel**, and V. Stein Foster. "Functionalizing ecological integrity: using functional ecology to monitor animal communities". *Frontiers in Ecology and the Environment*: e2852.

2025 – Ramírez Sánchez, D., G. Seingier, G. De León Girón, M. Villada Canela, **Z. L. Steel**, and H. Rivera Huerta. "Systematic review of the trends and methodologies in research on the effects of forest fires on avifauna in temperate forests". *Ardeola*: 72(1):105-120.

2024 – van den Bosch, M., J. Costanza, R. Peek, J. Mola, and **Z. L. Steel**. "Climate change scenarios forecast increased drought exposure for terrestrial vertebrates in the contiguous United States". *Communications Earth & Environment*: 708.

2024 – **Steel, Z. L.**, J. E. D. Miller, L. C. Ponisio, M. W. Tingley, K. Wilkin, R. Blakey, K. M. Hoffman, and G. Jones. "A roadmap for pyrodiversity science". *Journal of Biogeography* 51(2):280-293.

2023 – Francis, E. J., S. Pourmohammadi, **Z. Steel**, B. M. Collins, and M. D. Hurteau. "Proportion of forest area burned at high-severity increases with increasing forest cover and connectivity in western US watersheds". *Landscape Ecology*.

2023 – Stephens, S.[1], **Z. Steel**[1], B. Collins, D. Fry, S. J. Gill, H. Rivera-Huerta, and C. N. Skinner. "Climate and fire removal impacts on tree recruitment in mixed conifer forests in Northwestern Mexico and California". *Ecological Applications* 33(4):e2844.
[1]*Authors contributed equally to this work*

2023 – **Steel, Z. L.**, G. M. Jones, B. M. Collins, R. Green, A. Koltunov, K. L. Purcell, S. C. Sawyer, M. R. Slaton, S. L. Stephens, P. Stine, and C. Thompson. "Mega-disturbances cause rapid decline of

mature conifer forest habitat in California", *Ecological Applications 33(2):e2763*.

2023 – Weeks, J. M., H. Safford, J. Miller, **Z. Steel**, and E. Batzer. "High Severity Fire Drives Floristic Homogenization in Human-Altered Forests", *Ecosphere* 14(2):e4409.

2023 – Williams, J., H. Safford, N. Enstice, J. Tangenberg, **Z. Steel**, and A. Paulson. "High severity burn area and proportion exceed historic conditions in forests of Sierra Nevada and adjacent ranges (USA)". *Ecosphere* e4397.

2022 – Viljur, M. and 46 others including **Steel, Z. L.**. "The effect of natural disturbances on forest biodiversity: An ecological synthesis", *Biological Reviews* 97(5):1930-1947.

2022 – Levine, J., B. Collins, **Z. L. Steel**, P. de Valpine, and S. Stephens. "High-severity fire risk heightened both on and near industrially managed forests". *Frontiers in Ecology and the Environment* 20(7):397-404.

2022 – Safford, H., A. Paulson, **Z. Steel**, D. Young, and R. Wayman. "The 2020 California fire season: A year like no other, a return to the past, or a harbinger of the future?". *Global Ecology and Biogeography 31(10):2005-2025*.

2022 – **Steel, Z. L.**, A. Fogg, R. Burnett, L. J. Roberts, H. D. Safford. 2022. "When bigger isn't better – implications of large high-severity wildfire patches for avian diversity and community composition". *Diversity and Distributions* 23(3):439-453.

2021 – **Steel, Z. L.**, B. M. Collins, D. B. Sapsis, and S. L. Stephens. "Quantifying pyrodiversity and its drivers". *Proc. R. Soc. B* 288(1948).

2021 – **Steel, Z. L.**, M.J. Goodwin, M.D. Meyer, G.A. Fricker, H.S.J. Zald, M.D. Hurteau, and M.P. North. Do forest fuel reduction treatments confer resistance to beetle infestation and drought mortality? *Ecosphere* 12(1):e03344.

2021 – Jager, H., J. Long, R. Malison, B. Murphy, A. Rust, L. Silva, R. Sollman, **Z. Steel**, M. Bowen, J. Dunham, J. Ebersole, and R. Flitcroft. "Resilience of terrestrial and aquatic fauna to historical and future wildfire regimes in western North America". *Ecology and Evolution* 11(18):12259-12284.

2021 – Stephens, S., S. Thompson, G. Boisrame, B. Collins, L. Ponisio, K. Rakhmatulina, **Z. Steel**, J. Stevens, K. Wilken, and J. van Wagtendonk. "Fire, water, and biodiversity in the Sierra Nevada: A possible triple win". *Environmental Research Communications* 3(08):1004.

2021 – **Steel, Z. L.**, D. Foster, M. Coppoletta, J. M. Lydersen, B. Wing, S. L. Stephens, and B. M. Collins. "Ecological resilience and vegetation transition in the face of multiple large wildfires". *Journal of Ecology* 109(9):3340-3355.

2019 – **Steel, Z. L.**, B. Campos, W. Frick, R. Burnett, and H. D. Safford. "The effects of wildfire severity and pyrodiversity on bat occupancy and diversity in fire-suppressed forests". *Scientific Reports* 9, 16300.

2019 – Wilkins, L. G. E., K. R. Matthew, **Z. L. Steel**, S. C. Nussle, and S. M. Carlson. "Population dynamics of *Rana sierrae* at Dusy Basin: Influence of non-native predators, drought, and restoration potential". *Ecosphere* 10(11):e02951.

2018 – **Steel, Z. L.**, M. Koontz, and H. Safford. "The changing landscape of wildfire: Burn pattern trends and implications for California's yellow pine and mixed conifer forests". *Landscape Ecology* 33(7): 1159-1176.

2017 – **Steel, Z. L.**, A. E. Steel, J. Williams, J. H. Viers, P. Marquet, and O. Barbosa. "Patterns of bird diversity and habitat use in mixed vineyard-matorral landscapes of Central Chile". *Ecological Indicators 73:345-357*.

2017 – Grof-Tisza, P., **Z. L. Steel**, E. Cole, M. Holyoak, and R. Karban. "Testing predictions of movement behavior in a hilltopping moth". *Animal Behavioral* 133:161-168.

2017 – Grof-Tisza, P., **Z. L. Steel**, and R. Karban. "The spatial distribution and oviposition preference of the Ranchman's tiger moth, Platyprepia virginalis (Lepidoptera: Arctiidae)". *Journal of the Lepidopterists' Society* 71(1):16-19.

2015 – **Steel, Z. L.**, H. D. Safford, and J. H. Viers. "The fire frequency-severity relationship and the legacy of fire suppression in California forests". *Ecosphere* 6(1):1-23.

2013 – Schwartz, M.W., L. B. Smith, and **Z. L. Steel**. "Conservation investment for rare plants in urban environments". *PLOS ONE* 8(12):e83809.

2012 – **Steel, Z. L.**, M. L. Bond, R. B. Siegel, and P. Pyle. "Avifauna of Sierra Nevada Network parks: Assessing distribution, abundance, stressors, and conservation opportunities for 145 bird species". National Park Service, Fort Collins, Colorado. Natural Resource Report NPS/SIEN/NRR—2012/506.

2011 – Siegel, R. B, R. L. Wilkerson, J. F. Saracco, and **Z. L. Steel**. "Elevational distribution of common bird species on the Sierra Nevada's west slope". *Western Birds* 42:2-26.

In Review or Revision

Bolas, E. C., M. Dickie, E. Schoenborn, K. L. Calhoun, **Z. L. Steel**, E. G. Ritchie, R. M. Newsome, A. T. Ford, A. Sih, and J. A. Smith. "The synergistic effects of extreme fires and biodiversity threats on predator-prey interactions". *Biological Reviews*.

Steel, Z. L., H. M. Boone, L. McTigue, V. Stein Foster, M. van den Bosch, A. N. Stillman, C. D. Cappello, M. Ditmer, and E. J. Francis. "Range-wide effects of vegetation management on a rapidly declining woodland bird species". *Biological Conservation*.

Boone, H. M., M. van den Bosch, A. Stillman, E. Francis, M. Ditmer, **Z. L. Steel**. "Implications of active forest management for western U.S. birds". *Ecology Letters*.

Stillman, A. N., M. Stimas-Mackey, K. Norman, **Z. L. Steel**, S. Sawyer, M. W. Tingley, G. Duran, D. Fink, S. Ligocki, T. Auer, V. Ruiz-Gutierrez, and G. M. Jones. "Pyrodiversity drives spatially heterogeneous bird responses to fire across the western U.S." *Nature Ecology & Evolution*.

Calhoun, K. L., B. J. Furnass, A. Dheer, **Z. L. Steel**, C. E. Wilkinson, J. A. Smith, and M. W. Tingley. "Out of the fire and into the frying pan: Severe fire and exposure to the wildland-urban interface increase human-wildlife conflict". *Proceedings of the National Academy of Science*.

Parks, S., J. D. Coop, M. Coppoletta, **Z. L. Steel**, and C. Stevens-Rumann. "Forest persistence after repeated wildfire is compromised by first-entry moderate-severity fire". *Forest Ecology and Management*.

McTigue, L. E., and **Z. L. Steel**. "Mammal species show resilience to Megafires in the Southern Rocky Mountains". *Fire Ecology*.

Boone, H. M., R. Flitcroft, J. Goldberg, and **Z. L. Steel**. "Animal community sensitivities to anthropogenic threats across North America". *Nature Ecology and Evolution*.

Technical Reports & Software

2025 – van den Bosch, M., **Z. L. Steel**. "Projected changes in US vertebrate drought exposure". Prepared for California Department of Fish and Wildlife. RShiny application Available online

2024 – Eisenburg, C., S. Prichard, M. Nelson, P. Hessburg (lead authors) and 33 others including **Z. L. Steel**. "Braiding Indigenous and Western Knowledge for Climate-Adapted Forests: An Ecocultural State of Science Report". University of Washington. Available online

2021 – Meyer, M., Long, J., Safford, H. (editors) and 22 contributors including **Z. Steel**. "Postfire restoration framework for national forests in California". USDA Forest Service Pacific Southwest Research Station. PSW-GTR-270.

2021 – **Steel, Z. L.**, M. D. Meyer, M. P. North, A. Wuenschel, and S. M. Ostoja. "Reforestation tool for tree mortality landscapes." In: Meyer, M.D.; Long, J.W.; Safford, H.D., eds. Postfire restoration

framework for national forests in California. Gen. Tech. Rep. PSW-GTR-270. Albany, CA: U.S. Department of Agriculture, Forest Service, Pacific Southwest Research Station: 191-192. Appendix 7.

2021 – USFS Region 5 Climate Change Trend Assessments (8 reports). E.g.:

Wuenschel, A., S. Gross, B. Estes, K. Merriam, H. Safford, S. Sawyer, **Z. Steel**, L. Wolf. Sierra National Forest Climate Change Trend Summary. Unpublished report. USFS Pacific Southwest Region, Vallejo CA.

2020 – **Steel, Z. L.**, M. Meyer, A. Wuenschel, S. Ostoja, and M. North. "Climate-wise reforestation toolkit". Rshiny web application prepared for the US Forest Service, and the California Climate Hub. Available online

2018 – **Steel, Z. L.**, B. R. Campos, and H. D. Safford. "Bat occupancy in Sierra Nevada wildfire areas and implications for post-fire management". Prepared for US Forest Service, Region 5. PDF

2018 – Campos, B. R., **Z. L. Steel**. FIRE-BAT spatial predictive tool. ArcGIS toolbox prepared for US Forest Service, Region 5. Users Manual

2017 – Campos, B. R., R. D. Burnett, and **Z. L. Steel**. "Bird and bat inventories in the Storrie and Chips fire areas 2015-2016: Final report to the Lassen National Forest". Point Blue Conservation Science, Petaluma, CA. Point Blue Contribution No. 2142. PDF

2017 – Fogg, A. M, **Z. L. Steel** and R. D. Burnett. "Avian Monitoring in Freds and Power fires: Final Report". Point Blue Conservation Science, Petaluma, CA. Point Blue Contribution No. 2138. PDF

2013 – Schwartz, M. W., et al. "Sensitive Animals - Appendix 15 to SEKI Natural Resources Condition Assessment". Natural Resource Report NPS/SEKI/NRCA—2013/665. National Park Service. Fort Collins, Colorado. Assessments within:

Steel, Z. L., E. Reddy and R. Green. The American Pika - *Ochotona princeps*

Steel, Z. L., J. Shields. Sierra Nevada Bighorn Sheep - *Ovis canadensis sierrae*

Copeland, S., K. Dybala, **Z. L. Steel**. California Condor - *Gymnogyps californianus*

Steel, Z. L., Sierra Nevada Chipmunks - *Tamais spp.*

2012 – **Steel, Z. L.**, O. Barbosa, and P. Marquet. "Biodiversidad de Aves y Carbono en el Paisaje Vinedo". Three reports prepared for landowner partners in the Colchagua Valley, Chile.

2011 – **Steel, Z. L.**, M. Wilderson, P. Grof-Tisza, and K. Sulzner. "Assessing species and area vulnerability to climate change for the Oregon Conservation Strategy: Willamette Valley Ecoregion". Prepared for the Oregon Department of Fish and Wildlife and the Defenders of Wildlife.

SELECT PRESENTATIONS

Invited talks

2026 – Colorado State University, Department of Fish, Wildlife, and Conservation Biology Seminar. *Fire and Wildlife: A love-hate relationship*

2026 – American Fisheries Society. *Does fire ecology stop at the water's edge? Terrestrial-aquatic fire science opportunities*

2026 – The Nature Conservancey, Sacramento, California. *Applying fire research to address challenges for forest resilience and biodiversity in western conifer forests*

2025 – University of California, Davis, California Fire Science Consortium, FFERAL Lecture. *The search for general relationships between fire and biodiversity*

2025 – Boise State University, Department of Biological Sciences Seminar, *Wildfires and Wildlife: Conserving Biodiversity in an Era of Change*

2024 – North American Society for Bat Research, *Leveraging broad-scale monitoring to understand wildfire impacts on North American bats*

2024 – American Association for Ornithology, *Finding the shape of Pyrodiversity: Insights from a fire-restored landscape into ecological memory, scale, and the multi-dimensional response of biodiversity to varied fire histories*

2023 – University of Wisconsin Madison, Forest and Wildlife Ecology Seminar, *Wildlife in the Pyrocene: Understanding impacts of shifting fire regimes on wildlife communities*

2023 – Journal of Fire Ecology, *Effects of wildfire and prescribed fire on wildlife panel*

2022 – California Native Plant Society Conference, *Declining southern Sierra Nevada conifer forests in an age of mega-disturbances: applying remote sensing tools to inform mature forest conservation*

2021 – International Fire Ecology and Management Congress, *Forest resilience and transition in the face of successive large wildfires in California's Sierra Nevada Mountains*

2021 – Berkeley EcoLunch Seminar, *Wildlife and wildfire: a love-hate relationship*

2021 – Yosemite National Park Forum, *Restoring pyrodiversity to benefit Sierra Nevada wildlife communities*

2021 – Western Section of the Wildlife Society Annual Meeting **(Plenary)**, *Implications of changing fire patterns on habitats and wildlife*

2020 – Natural Areas Association Annual Meeting, *Implications of changing fire regimes for Sierra Nevada bat and bird communities*

2017 – Instituto Politécnico Nacional, CIIDIR-Unidad Oaxaca, Mexico, *Murcielagos en un paisaje caracterizado por incendios: Impactos en diversidad y uso de habitat*

2017 – Northwest Science Conference, *The influence of burn severity on bat species occurrence in post-fire landscapes*

2011 – Universidad Mayor, Santiago, Chile, *Mercados de carbono: una oportunidad potencial para biodiversidad en agroecosistemas*

Contributed talks

2026 – American Ornithological Society annual meeting, **Steel, Z. L.**, H. M. Boone, L. McTigue, V. Stein Foster, M. van den Bosch, A. N. Stillman, C. D. Cappello, M. Ditmer, and E. J. Francis. *Effects of active forest management on the declining pinyon jay (Gymnorhinus cyanocephalus)*

2023 – International Association of Landscape Ecology, *Burning for biodiversity: Finding pyrodiversity optimums for multiple taxa in a managed wildfire area*

2022 – Yosemite National Park Hydroclimate Meeting, *The chiropteran fire pulse: vegetation and hydro-climate mediate changes in bat activity in Sierra Nevada wilderness*

2022 – International Association of Landscape Ecology, *Megadisturbances Cause Rapid Decline of Mature Forest Habitat in California*

2021 – International Association of Landscape Ecology, *Quantifying pyrodiversity and its drivers*

2020 – Ecological Society of America Annual Meeting, *What doesn't kill you makes you stronger - or not: Conifer mortality following forest treatments and drought*

2019 – The Wildlife Society Annual Conference, *The Influence of Burn Severity and Pyrodiversity on Bat Communities in Sierra Nevada Forests*

2018 – Mediterranean Forest Ecosystems International Conference (MedPine6), *Changing patterns of severe wildfire lead to bird diversity declines & community transitions in conifer forests of California, USA*

2018 – Ecological Society of America Annual Meeting, *Bat species occupancy varies across a disturbance gradient; implications for a future of extreme wildfire events*

2017 – Ecological Society of America Annual Meeting, *The influence of spatial context and successional pathways on bird communities following high-severity wildfire*

2017 – Mediterranean Ecosystems Conference (MEDECOS), *Shifting patterns of California fire and forest landscapes in an era of global change*

2016 – Natural Areas Conference, *The Changing Landscape of California Fire: Trends in Burn Patterns and Post-Fire Forest Heterogeneity*

2011 – Buenos Aires Fulbright Regional Enhancement Seminar, *Carbon Sequestration: Risk or tool for biodiversity?*

2011 – Bay Area Conservation Biology Symposium, *Assessing species vulnerability to climate change in the Willamette Valley Ecoregion*